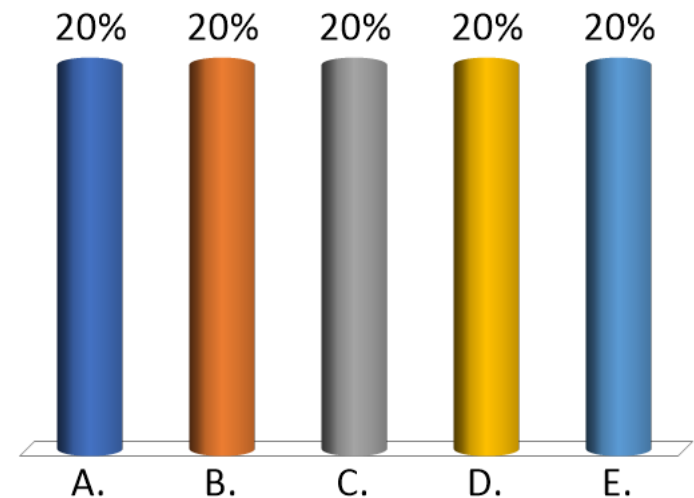


BPM501
ER Module
IMCQ 3
Spring 2026 - KEY

1. A 35-year-old woman comes to the emergency department because of severe, sudden onset of right lower quadrant pain. The pain began during exercise and radiates towards her groin. She also reports nausea, vomiting, and a fainting episode. Her periods have been regular. Physical examination shows right lower quadrant abdominal tenderness-with some guarding. Pelvic ultrasonography shows an enlarged, edematous right ovary with decreased blood flow. She is diagnosed with is ovarian torsion. Which of the following ligaments is most likely associated with this condition?

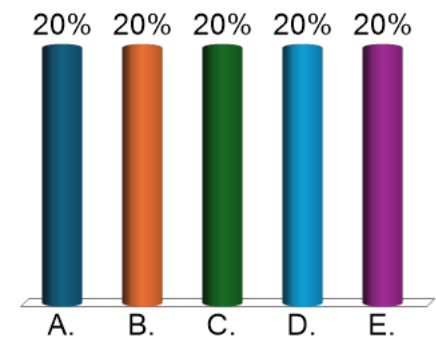
- A. Suspensory
- B. Broad
- C. Uterosacral
- D. Cardinal
- E. Round



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- D. Cardinal
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SOM.MKII.BPM2.1.ER.1.ANAT.1022: Describe the ovaries and ovarian ligaments.

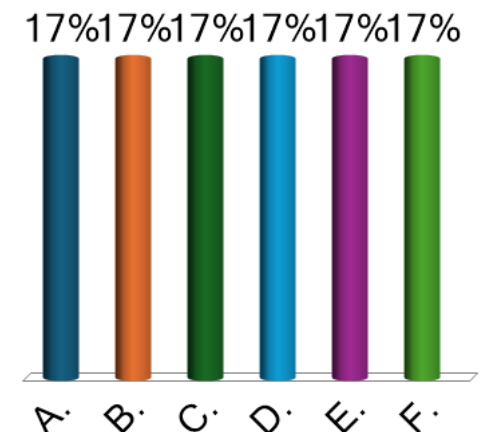


2. A 23-year-old woman comes to the physician because of a 5-day history of swelling and pain in her genital area, accompanied by low-grade fever and difficulty walking. Her mother notes that the swelling is located near the vaginal opening and has gradually become more tender. Physical examination shows a fluctuant, erythematous mass at the lower left vaginal introitus, medial to the labia majora. The overlying skin is warm and tender. An infected Bartholin gland cyst is suspected. Which of the following combinations most accurately identifies the nerve responsible for pain sensation in the affected area and the structures responsible for its blood supply and lymphatic drainage?

- A. Anterior labial nerves, external pudendal artery, superficial inguinal nodes
- B. Inferior rectal nerve, inferior rectal artery, iliac nodes
- C. Dorsal nerve of the clitoris, deep artery of the clitoris, deep inguinal nodes
- D. Posterior labial nerves, perineal artery, superficial inguinal nodes
- E. Cavernous nerves, perineal artery, deep inguinal nodes
- F. Genitofemoral nerves, femoral artery, deep inguinal nodes

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- F. Genitofemoral nerves, femoral artery, deep inguinal nodes



SOM.MKII.BPM2.1.ER.1.ANAT.1014: Describe the course of the pudendal nerve and its branches within the female perineum.

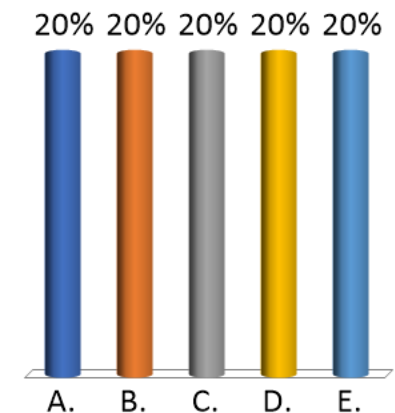
3. A 38-year-old man comes to the physician because of a 1-week history of dull, aching pain in the perineum. He denies fever but reports mild discomfort while walking and sitting. He recently completed a long road trip, driving for several hours daily. Physical examination shows fullness and tenderness on the left side of the perianal area. An MRI of his perineum is shown. Which of the following structures is most likely to form the lateral boundary of the affected space?

- A. Obturator internus muscle
- B. External anal sphincter
- C. Levator ani muscle
- D. Sacrotuberous ligament
- E. Gluteus maximus muscle



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SOM.MKII.BPM2.1.ER.1.ANAT.1008: Describe the ischioanal fossa, its location, contents, innervation and clinical significance.

4. A 58-year-old woman comes to the physician because of a 3-month history of abnormal vaginal bleeding. She describes the bleeding as intermittent spotting, sometimes heavier than a normal period. Physical examination shows a small amount of dark red blood in the vaginal vault. Microscopic examination of an endometrial biopsy specimen shows crowded endometrial glands with irregular shapes and branching pattern, lined by stratified columnar epithelial cells. Which of the following best describes a normal feature of the structure from which the biopsy specimen was taken?

- A. Lined by a simple columnar epithelium
- B. Interlacing bands of smooth muscle cells
- C. Lined by a simple squamous epithelium
- D. Epithelial cells are filled with lipid droplets for hormone production
- E. Composed of abundant adipose tissue and terminal ductules

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5. A 45-year-old woman comes to the physician because of a 2-week history of a persistent pelvic pain and a 3-month history of irregular menstrual cycles. Ultrasonography of the abdomen shows a mass on the right ovary. Examination of a biopsy specimen obtained from the ovarian mass shows morphological changes of the cells highlighted by the yellow arrow in the photomicrograph of a normal biopsy shown. Which of the following is most likely a function of a common tissue normally influenced by the hormones produced by the affected cell type?

- A. Provides an environment for fetal development
- B. Production of testosterone by the adrenals
- C. Promotes development into a mature egg
- D. Stimulates secretion of progesterone from the placenta
- E. Stimulates development of the germinal epithelium



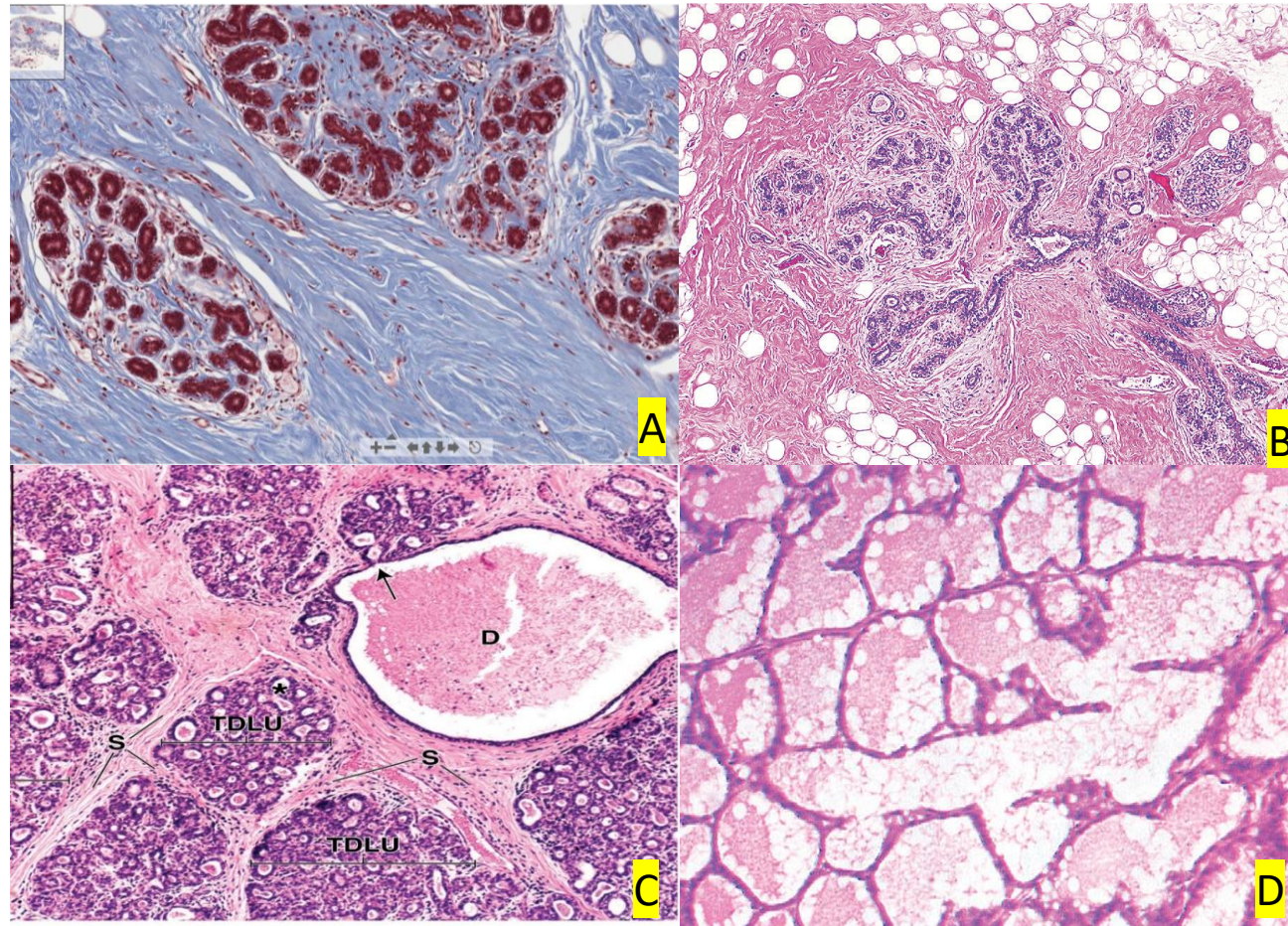
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- C. Promotes development into a mature egg
- D. Stimulates secretion of progesterone from the placenta
- E. Stimulates development of the germinal epithelium



6. A 32-year-old woman comes to the physician because of a 2-day history of a lump in her right breast. She is 6 weeks postpartum, and is breast feeding. Physical examination shows a well-defined, hypermobile 2.5-cm x 2.5-cm mass. A biopsy specimen of the mass shows normal breast tissue in an actively lactating mammary gland. Which of the following photomicrographs best identifies the mammary gland at this stage?

- A. A
- B. B
- C. C
- D. D



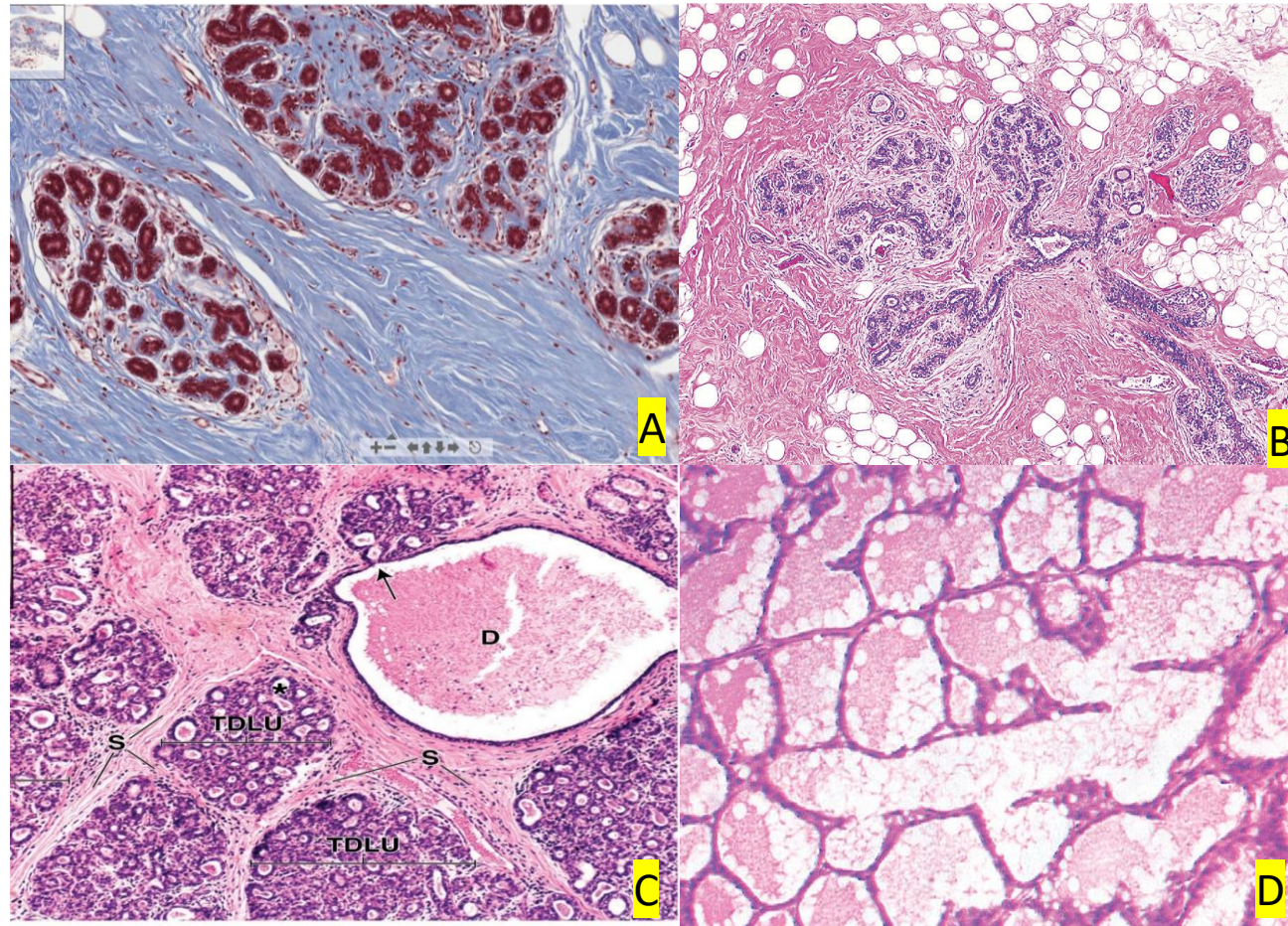
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A. A

B. B

C. C

D. D



SOM.MK.I.BPM2.2.ER.1.HCB.1176 Describe effects of hormones on the mammary glands during lactation.

7. A 5-day-old boy is brought to the emergency department by his mother because of a generalized tonic-clonic seizure. He was born at home at 39-weeks gestation, and his mother did not have any prenatal care. His weight and height are at the 30th percentile, and head circumference is at the 2nd percentile. Physical examination shows a fair-skinned infant with blue eyes, and a musty body odor. Laboratory studies show elevated serum phenylalanine levels. Cerebrospinal fluid analysis shows decreased dopamine and serotonin metabolites. Which of the following coenzymes is most likely deficient in this patient?

- | | |
|--------------------------------|----|
| A. Iodide | 0% |
| B. Vitamin B12 | 0% |
| C. Tetrahydrofolate | 0% |
| D. Tetrahydrobiopterin | 0% |
| E. Vitamin C | 0% |
| F. S-adenosyl methionine (SAM) | 0% |



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SOM.MK.II.BPM2.1.ER.2.BCHM.1014; Highlight the use of tetrahydrobiopterin (BH4), PLP, vitamin C (ascorbic acid), and S-adenosyl methionine in the synthesis of Dopamine, Norepinephrine and Epinephrine. Identify the various enzymes required.



8. A 15-day-old boy is brought to the emergency department by his mother because of a 2-day history of vomiting, lethargy, and poor feeding. His pulse is 126/min and blood pressure is 65/40 mmHg. He is irritable and has a weak cry. Physical examination shows sunken fontanelle and feeble pulses. Laboratory studies are shown in the table. Deficiency of which of the following enzymes best explains the development of the patient's condition?

Parameter (Serum)	Patient	Reference
Glucose	30	50-70mg/dL
Sodium (Na ⁺)	125	133-146 mmol/L
Potassium (K ⁺)	5.7	3.5-5.3 mmol/L
Dehydroepiandrosterone (DHEA)	4.5	<1.0 mmol/L
Cortisol	50	138-635 nmol/L
17-hydroxyprogesterone	150	10-48 ng/dL

- A. Desmolase

0%
- B. Aldosterone synthase (CYP 11B2)

0%
- C. 17-beta hydroxylase (CYP 17)

0%
- D. Aromatase (CYP 19)

0%
- E. 21-alpha hydroxylase (CYP 21)

0%
- F. 11-beta hydroxylase (CYP 11B1)

0%
- G. 3-beta hydroxysteroid dehydrogenase

0%



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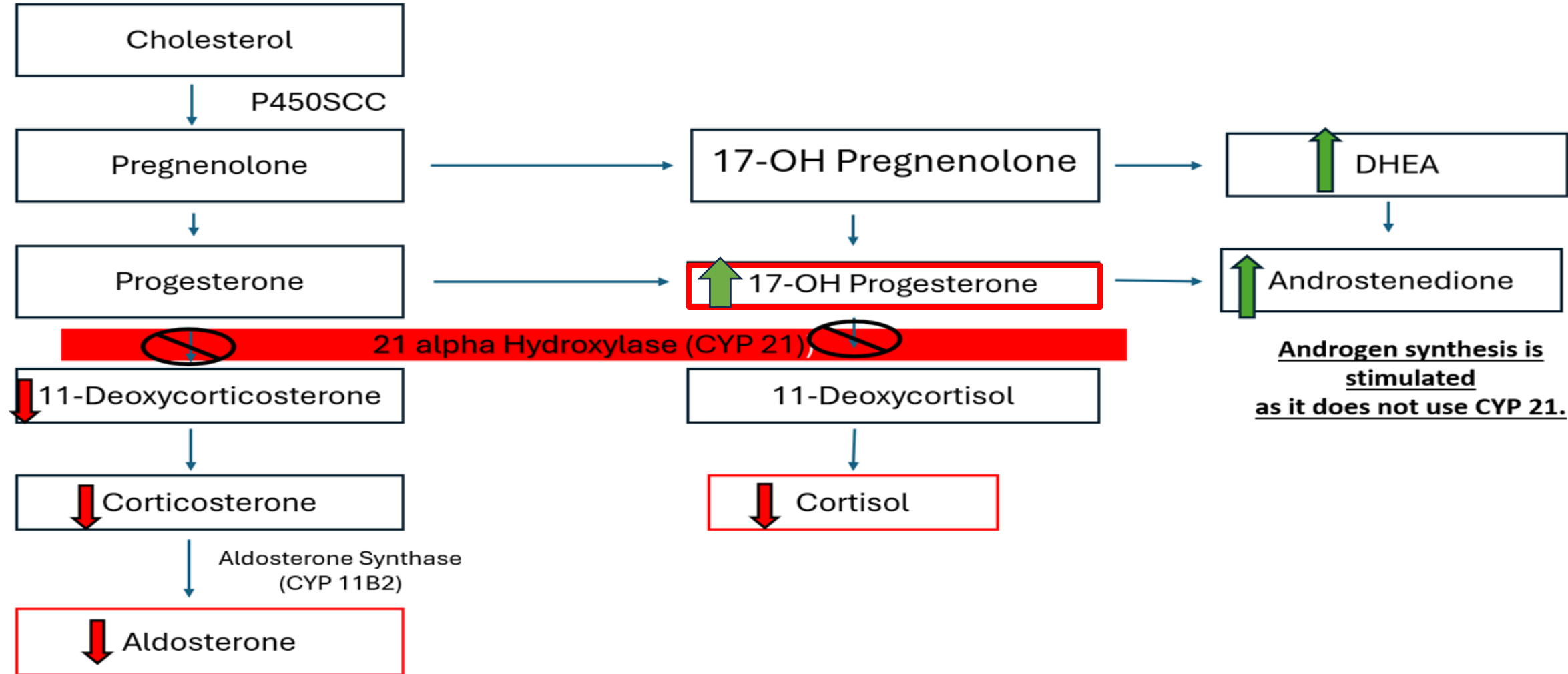
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- C. 17-beta hydroxylase (CYP 17)
- D. Aromatase (CYP 19)
- ✓ E. 21-alpha hydroxylase (CYP 21)
- F. 11-beta hydroxylase (CYP 11B1)
- G. 3-beta hydroxysteroid dehydrogenase

SOM.MK.II.BPM2.1.ER.2.BCHM.1030; Discuss congenital adrenal hyperplasias (CAH) and describe clinical features and biochemical basis of manifestations in patients with deficiencies of 3-hydroxysteroid DH, CYP 21, CYP 11B and CYP 17. Differentiate between the various enzyme deficiencies resulting in CAH.

CAH: Deficiency of CYP 21 (21-Hydroxylase)

Most common



9. A 55-year-old man comes to the physician because of a 3-month history of recurrent watery diarrhea, episodes of facial flushing, and intermittent diarrhea. Physical examination shows a holosystolic murmur heard best along the left lower sternal border. Echocardiography shows thickened tricuspid valve leaflets with severe regurgitation. A 24-hour urine collection shows markedly elevated 5-hydroxyindoleacetic acid (5-HIAA) excretion. Which of the following enzymes is most likely directly responsible for producing the urinary metabolite measured in this patient?

- | | |
|--|----|
| A. Tryptophan hydroxylase | 0% |
| B. DOPA decarboxylase | 0% |
| C. Phenylethanolamine N-methyl transferase | 0% |
| D. 5'-Deiodinase | 0% |
| E. 17-hydroxylase (CYP 17) | 0% |
| F. Monoamine oxidase | 0% |
| G. Catechol O-methyl transferase | 0% |

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SOM.MK.II.BPM2.1.ER.2.BCHM.1021 Explain the formation of 5-HIAA formation (Role of MAO) and analyze its clinical significance

10. A 16-year-old girl is brought to the emergency department by her father because of a 2-day history of nausea, increased urination, and vomiting. She has lost 2 kg (4.2 lb) over the past two weeks. Her blood pressure is 70/90 mmHg. Laboratory studies on admission show:

Random plasma glucose: 340 mg/dL (70-100mg/dL)

Glycated hemoglobin: 7.2% (<6%)

Urinalysis: Positive for glucose and ketones

Which of the following additional findings is most likely in this patient?

- | | |
|--|----|
| A. Target tissue insulin resistance | 0% |
| B. Low T4 and T3; Elevated TSH levels | 0% |
| C. Elevated C-peptide and proinsulin levels | 0% |
| D. Pancreatic islet cell autoantibodies | 0% |
| E. Low serum cortisol levels and elevated ACTH | 0% |
| F. Elevated urinary homovanillic acid (HVA) levels | 0% |

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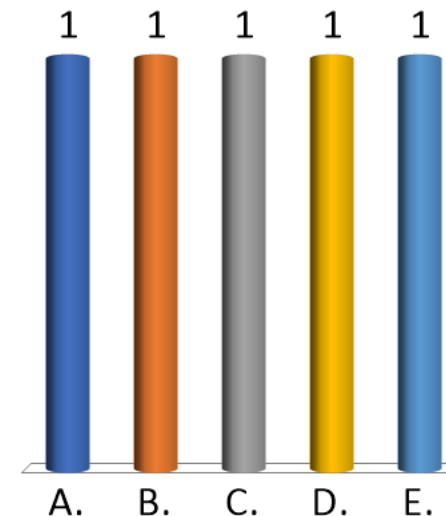
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- E. Low serum cortisol levels and elevated ACTH
- F. Elevated urinary homovanillic acid (HVA) levels

SOM.MK.II.BPM2.1.ER.2.BCHM.1034; Distinguish between the types of diabetes: type 1 and type 2 (differences in insulin secretion) and secondary diabetes mellitus.

11. A 28-year-old woman comes to the physician because of a 2-month history of failure to lactate, amenorrhea, cold intolerance, fatigue, and nausea. Two months ago, she delivered a female newborn through vaginal delivery, but her labor was complicated by a heavy blood loss. Her temperature is 36.4°C (97.5°F), pulse is 58/min, and blood pressure is 110/70 mm Hg. Physical examination shows dry mucous membranes, pale tongue and conjunctiva, and scanty pubic and axillary hair. Genital examination shows vaginal atrophy and dryness. Serum studies show low concentrations of all anterior pituitary hormones. A peripheral blood smear shows normocytic normochromic anemia. MRI of the brain shows atrophy of the pituitary gland. Secretion of which of the following hormones is most likely increased in this patient?

- A. Estrogen
- B. Cortisol
- C. T3 and T4
- D. Prolactin
- E. Thyrotropin releasing hormone (TRH)



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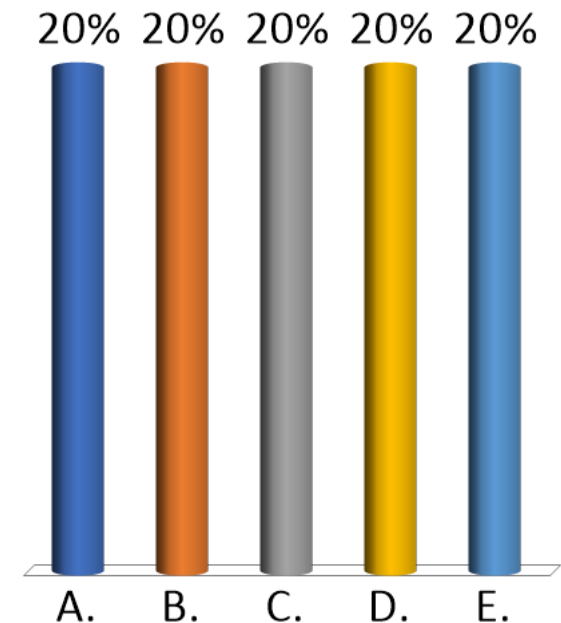
- A. Estrogen
- B. Cortisol
- C. T3 and T4
- D. Prolactin
- ✓ E. Thyrotropin releasing hormone (TRH)

SOM.MK.II.BPM2.1.ER.1.PHYS.1014. Describe the effects of loss of all pituitary hormones and effects of injury to pituitary stalk.



12. A 51-year-old man comes to the physician because of a 7-month history of 5-kg (11-lb) weight gain and recurrent episodes of headaches. His blood pressure is 172/94 mm Hg; other vital signs are within normal limits. Physical examination shows thin extremities, abdominal striae, a non-healing wound in the left lower limb, and increased fat deposition in his trunk, face, and the back of the neck. Laboratory studies show increased serum concentrations of glucose, cortisol, and adrenocorticotrophic hormone (ACTH). Skeletal radiography shows osteoporosis. He is diagnosed with Cushing syndrome. Which of the following mechanisms best explains hypertension in this patient?

- A. Increased aldosterone action on renal tubular cells
- B. Increased insulin sensitivity
- C. Increased sensitivity of angiotensin II and norepinephrine
- D. Activation of renin-angiotensin-aldosterone system
- E. Mineralocorticoid-like effect of the adrenal androgen

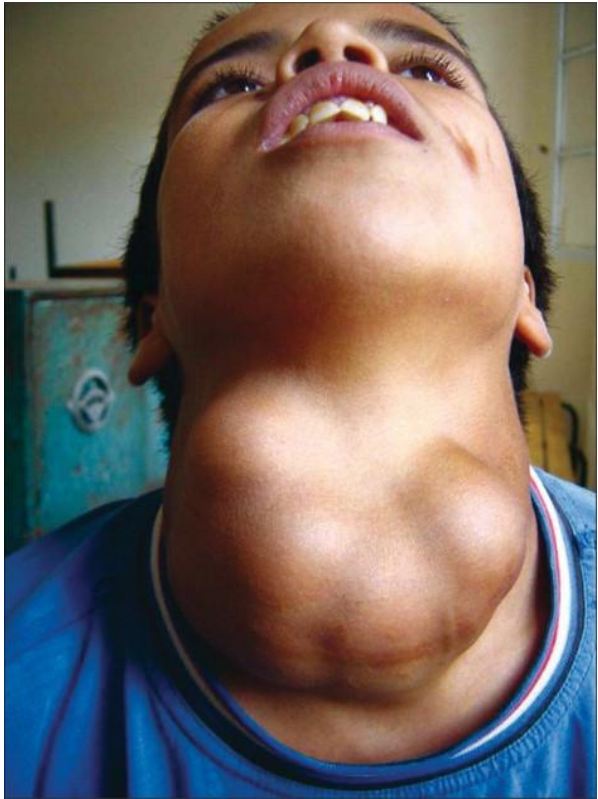


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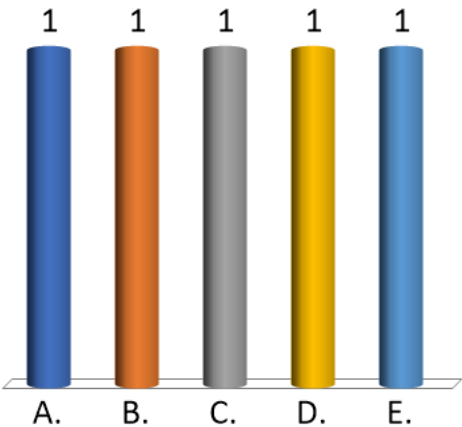
SOM.MK.II.BPM2.1.ER.1.PHYS.1055. List and explain the causes of thyroid gland enlargement. Predict the consequences of thyroid hormone under- and over-secretion.

13. A 12-year-old boy is brought to the physician by his mother because of a 1-year history of swelling in the front of the neck, lethargy, constipation, and a 5-kg (11-lb) weight gain. His temperature is 36.5°C (97.7°F), pulse is 56/min, and blood pressure is 122/84 mm Hg. Physical examination shows dry skin, sluggish deep tendon reflexes, and an enlarged thyroid (goiter) as shown. Laboratory studies show elevated serum thyroglobulin concentration and absence of autoantibodies. A 24-hour urinary iodine excretion is significantly low. Which of the following sets of alterations in thyroid function study is most likely in this patient?

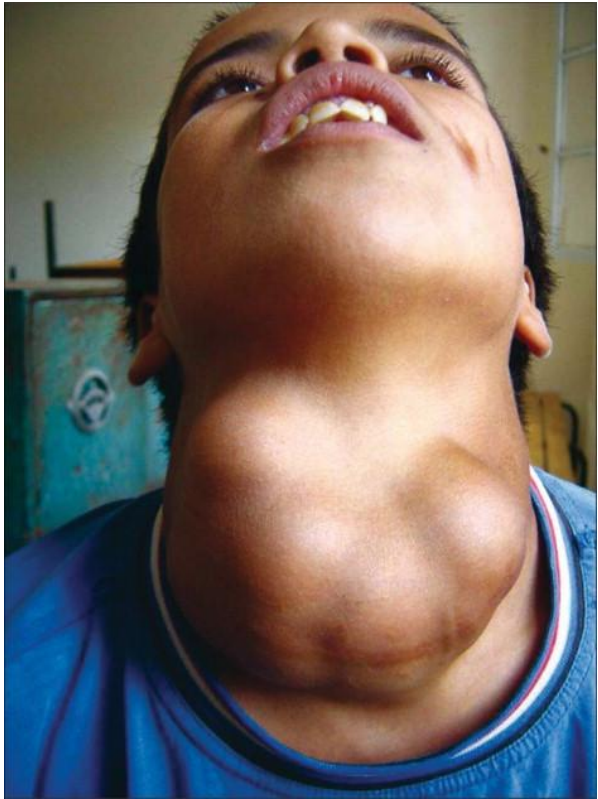


Option	Serum free T4	Serum TSH	BMR
A	↓	↓	↓
B	↓	↑	↑
C	↓	↑	↓
D	↑	↓	↑
E	↑	↓	↓

(BMR=Basal metabolic rate)



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Option	Serum free T4	Serum TSH	BMR
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B	↓	↑	↑
✓ C	↓	↑	↓
D	↑	↓	↑
E	↑	↓	↓

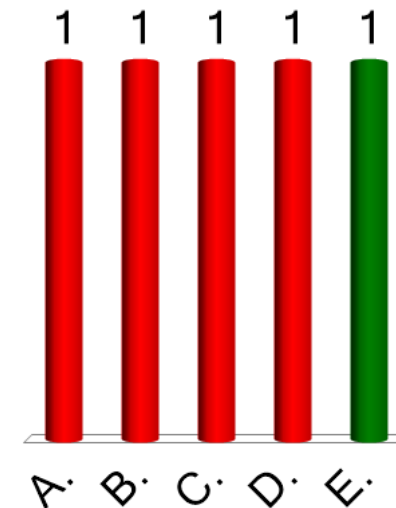
(BMR=Basal metabolic rate)

SOM.MK.II.BPM2.1.ER.1.PHYS.1055. List and explain the causes of thyroid gland enlargement. Predict the consequences of thyroid hormone under- and over-secretion.



14. A 39-year-old woman comes to the physician because of a 6-week history of constipation, increased urination, increased thirst, heartburn, and abdominal pain. Nine months ago, she underwent surgery to remove kidney stones. Vital signs are within normal limits. Physical examination shows no abnormalities. Serum studies show elevated calcium, parathyroid hormone (PTH), and normal glucose, sodium and potassium concentrations are within reference ranges. Ultrasonography of the neck shows a mass in the left parathyroid region. Radiography of long bones show lytic lesions and subperiosteal osteoporosis. Which of the following additional diagnostic findings is most likely in this patient?

- A. Long QT interval in ECG
- B. Elevated serum HbA1C concentration
- C. Reduced serum vasopressin concentration
- D. Hypertonic urine
- E. Increased urinary phosphate



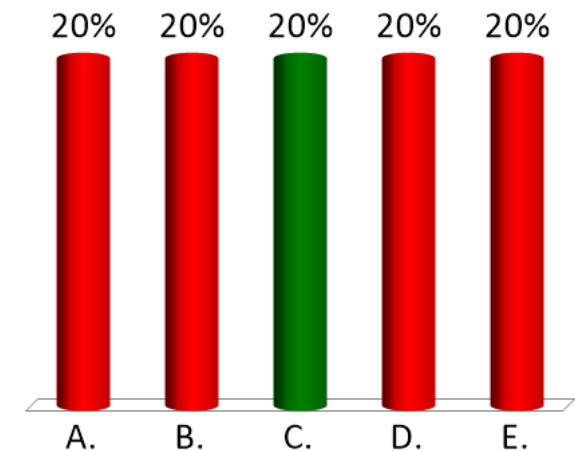
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- C. Reduced serum vasopressin concentration
- D. Hypertonic urine
- ✓ E. Increased urinary phosphate

SOM.MK.II.BPM2.1.ER.1.PHYS.1063. Explain the clinically relevant consequences of oversecretion/undersecretion of parathyroid hormone.

15. A 26-year-old man comes to the physician because of a 2-year history of infertility. He is married three years ago, and the couple has decided to have children. He has no history of difficulty erection. He has no chronic illness and takes no medications. Vital signs are within normal limits. Physical examination shows normal external genitalia and testicular volume. Serum testosterone, LH and FSH concentrations are within the reference ranges. Doppler studies show normal blood flow in the abdominal and pelvic blood vessels. Semen studies show low sperm count, normal sperm motility and morphology, and absent androgen-binding protein. Which of the following cell type dysfunction is the most likely cause of infertility in this patient?

- A. Spermatogonia of the seminiferous tubules
- B. Leydig cells of the testis
- C. Sertoli cells of the seminiferous tubules
- D. Zona reticularis cells of the adrenal cortex
- E. Gonadotrophs of the anterior pituitary



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- ✓ C. Sertoli cells of the seminiferous tubules
- D. Zona reticularis cells of the adrenal cortex
- E. Gonadotrophs of the anterior pituitary

SOM.MK.II.BPM2.1.ER.1.PHYS.1071 Describe the neural, vascular, and endocrine components of the erection and ejaculation response.

16. A 24-year-old woman comes to the physician for a routine examination. She says she has decided to marry and wants to check her fertility. Menarche occurred at 13 years of age. Her menses occur at regular 28-day intervals and last for 3-5 days. She has no chronic illness. Vital signs are within normal limits. Physical examination shows normal external genitalia, normal uterus, and normal breast development. Serum hormone studies are within the reference ranges. Microscopic examination of the cervical mucus shows maximum ferning pattern. She says she is on 12th day of her cycle today. Which of the following sets of physiological actions of estrogen is most likely in this patient currently?

Option	On LH secretion	On endometrium	On mammary gland
A	Positive feedback	Secretory changes	Ductal proliferation
B	Negative feedback	Secretory changes	Ductal proliferation
C	Positive feedback	Proliferative changes	Glandular proliferation
D	Negative feedback	Proliferative changes	Glandular proliferation
E	Positive feedback	Proliferative changes	Ductal proliferation
F	Negative feedback	Secretory changes	Glandular proliferation

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